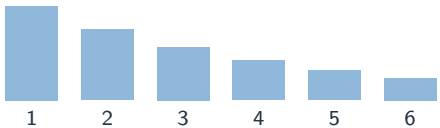
Условие

Нека X има пораждаща функция $g_X(s) = \frac{s}{4-3s}$. Намерете $\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathbb{P}(X = 1)$, $\mathbb{P}(X = 2)$.

Развиваме в ред

$$g_X(s) = \frac{s}{4} \cdot \frac{1}{1 - \frac{3}{4}s} = \frac{s}{4}\left(1 + \frac{3}{4}s + \frac{9}{16}s^2 + \dots\right) = \frac{1}{4}s + \frac{3}{16}s^2 + \dots$$

Коефициентът пред s^k е $\mathbb{P}(X = k)$.



Производни в 1

$g'(s) = \dfrac{4}{(4-3s)^2}$ и $g''(s) = \dfrac{24}{(4-3s)^3}$, откъдето $g'(1) = 4$, $g''(1) = 24$.

$$\mathrm{Var}(X) = g''(1) + g'(1) - (g'(1))^2 = 24 + 4 - 16 = 12.$$

Отговор. $\mathbb{E}[X] = 4$, $\mathrm{Var}(X) = 12$, $\mathbb{P}(X = 1) = \frac{1}{4}$, $\mathbb{P}(X = 2) = \frac{3}{16}$.